# Erfan Amani Bani

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## Education

**London Business School** — London, UK
*Ph.D. in Management Science and Operations* — *2025 – 2030 (Expected)*

**Sharif University of Technology** — Tehran, Iran
*M.Sc. in Industrial Engineering (Optimization Methods)* — *2023 – 2025*

**Sharif University of Technology** — Tehran, Iran
*B.Sc. in Industrial Engineering* — *2019 – 2023*

## Research Interests

**Methodology:** Optimization, machine learning.
**Applications:** Healthcare operations, sustainable operations.

## Selected Publications

[1] **Amani Bani, E.**, & Eshghi, K. (2026). *Quantum modeling of the dynamic ride-sharing problem: Development of quantum Benders decomposition methods*. Expert Systems with Applications, 310, 131320. [DOI]

[2] Fallahi, A., **Amani Bani, E.**, & Varmazyar, M. (2025). *Towards sustainable scheduling of unrelated parallel batch processors: A multi-objective approach with triple bottom line, classical and data-driven robust optimization*. Computers & Operations Research, 173, 106863. [DOI]

[3] **Amani Bani, E.**, Fallahi, A., Varmazyar, M., & Fathi, M. (2022). *Designing a sustainable reverse supply chain network for COVID-19 vaccine waste under uncertainty*. Computers & Industrial Engineering, 174, 108808. [DOI]

[4] Fallahi, A., **Amani Bani, E.**, & Niaki, S. T. A. (2022). *A constrained multi-item EOQ inventory model for reusable items: Reinforcement learning-based differential evolution and particle swarm optimization*. Expert Systems with Applications, 207, 118018. [DOI]

## Professional Experience

**Snapp!** — Mar. 2025 – Sep. 2025
*Senior AI Engineer* (Largest ride-hailing and delivery platform in the Middle East) *Tehran, Iran*

- Engineered low-latency AI pipelines for large-scale customer service operations.
- Built classification models to detect multihoming drivers, supporting platform-loyalty analytics.
- Designed ML-based behavioural prediction models informing dynamic incentive policies across a network of 2M+ drivers.

**FLYTZI** Feb. 2024 – Feb. 2025
*Co-founder & Optimization Engineer* *Tehran, Iran*

- Co-founded a mid-mile logistics platform matching vendors with drivers for parcel delivery.
- Formulated and solved the underlying capacitated vehicle routing problem with time windows; developed metaheuristic solvers for real-time dispatch at operational scale.
- Designed a 3D bin-packing module for parcel-to-vehicle assignment, integrating geometric feasibility with routing decisions.

## Teaching Experience

*Sharif University of Technology*

- **Teaching Assistant:** Statistics; Engineering Economics; Plant Layout; Project Control; Operations Research; Probability Theory; Simulation.

*London Business School*

- **Teaching Assistant:** Data Analytics for Finance.

## Academic Service

**Journal Reviewer:** *Transportation Research Part E*; *Expert Systems with Applications*; *IEEE Transactions on Pattern Analysis and Machine Intelligence*.

## Volunteering

**Gamein — Nationwide Gamified Education Event** 2022 – 2023
*Game Design and Development Team Lead* *Sharif University of Technology*

- Led product development for a national educational simulation reaching 3,300+ participants across Iranian universities.
- Coordinated a cross-functional team spanning game design, software engineering, and operations modelling.

## Honors & Awards

- LBS PhD Fellowship, full funding, London Business School, 2025–2030.
- Direct admission to M.Sc. programme (Talented Student Quota), Sharif University of Technology, 2023.
- Top 0.2% nationally, Iranian University Entrance Exam (Konkur), among 150,000+ candidates, 2019.

Last updated: Oct, 2026